Code parallelization

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Code parallelization

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- How to use the parallelized version of PALM
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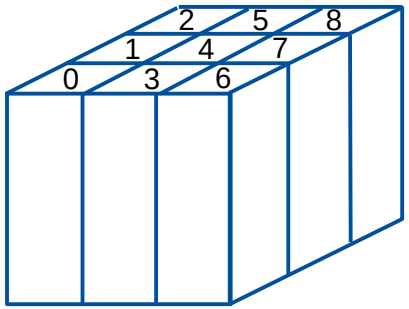
Code parallelization

Basics of parallelization

- All processor elements (PE, core) carry out the same program code (SIMD)
- Each PE of a computer operates on a different set of data

- **Realization:**
  - Each PE solves the equations for a different subdomain of the total domain.



  - Program loops are parallelized using directives. Each PE solves for a subset of the total index range.

```
!$OMP DO
    DO  i = 1, 100
        .
        .
    ENDDO
```

  - Each PE only knows the variables' values of it's subdomain.
  - Communication / data exchange between PEs required.
  - Done by **M**essage **P**assing **I**nterface (MPI).

  - Parallelization can be easily done by the compiler, if all PEs have access to all variables (shared memory).
  - Shared memory model (OpenMP), also used for GPU programming (OpenACC).

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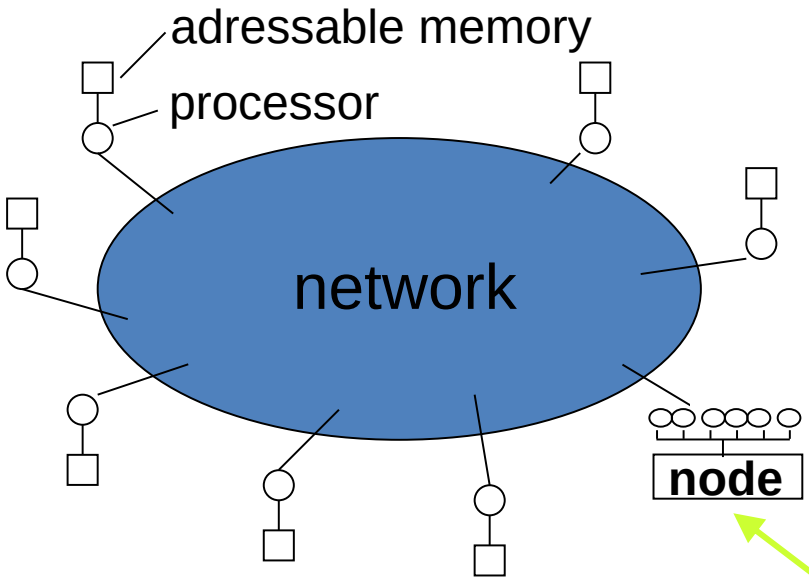
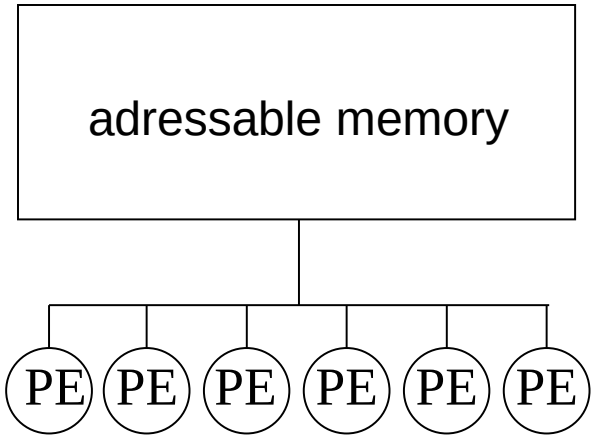
Basic architectures of massively parallel computers



distributed memory
(Cray-XC40)



shared memory
(manycore PCs)

clustered hybrid systems

(Linux-Cluster,
Bull/Atos, Cray-XC, NEC-Aurora)

MPI

OpenMP

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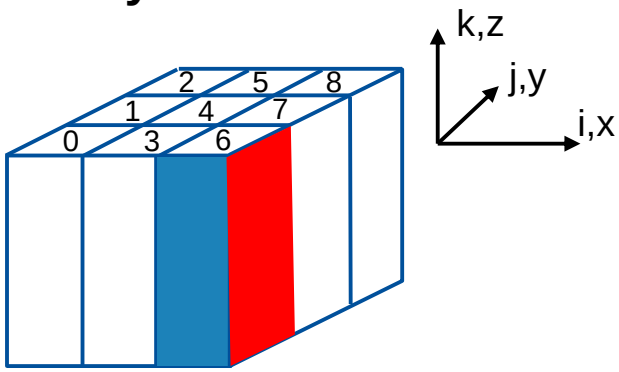
PALM parallelization concept

- **General demands for a parallelized program:**
  - **Good load balancing** (all cores should have same workload)
  - **Small communication overhead** (otherwise speed-up may get lost)
  - **Scalability** (up to large number of processors, otherwise they can‘t be used)
- **Basic parallelization method used for PALM is a 2D domain decomposition along x and y:**



Data to be communicated should be contiguous in memory (Fortran)!

f(i,j,k)
- columns of i , no contiguous data at all

f(k,j,i)
- columns of k , planes of k,j (all data contiguous)

  - No vertical decomposition because calculations for surface grid points are more expensive → No load balance!
  - Array indexing f(k,j,i) is used in PALM.
  - Message passing is realized using MPI.
  - OpenMP parallelization as well as hybrid usage of OpenMP and MPI is realized.

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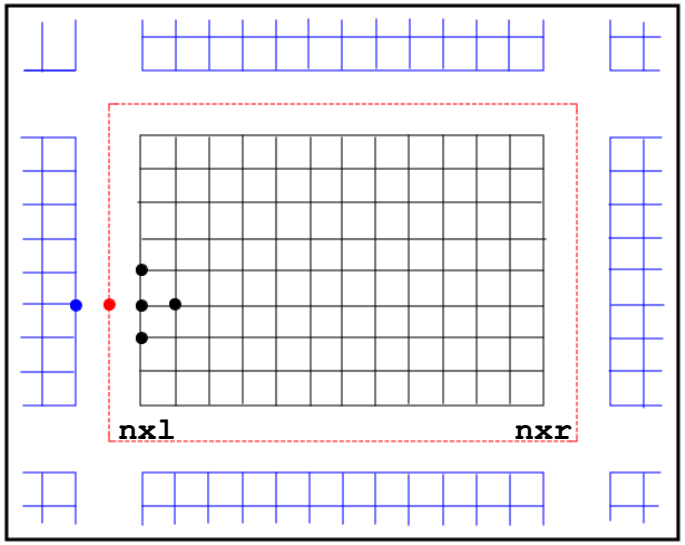
PALM parallelization model

- Central finite differences cause local data dependencies (only boundary data have to be transferred)

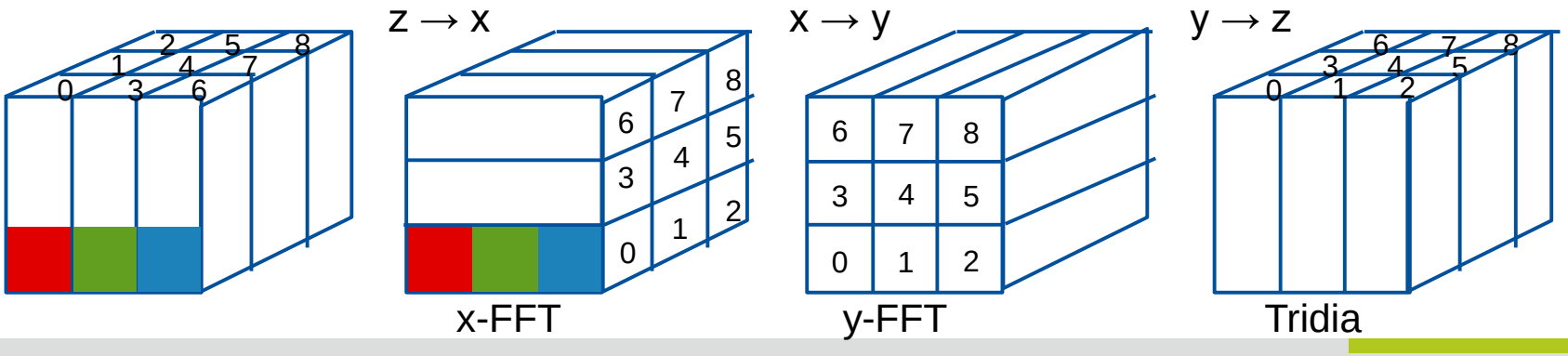
$$\left.\frac{\partial \psi}{\partial x}\right|_i = \frac{\psi_{i+1} - \psi_{i-1}}{2\Delta x}$$

  - **Solution:** introduction of ghost points
    `u(:,:,nxl:nxr), u(:,:,nxl-1:nxr+1)`



- FFT and linear equation solver cause non-local data dependencies (all 3D data have to be transferred)
  - **Solution:** transposition of 3D arrays

Example below shows transpositions for solving the Poisson equation.



Code parallelization

How to use the parallelized version of PALM

- The parallel version of PALM is activated by line

  ```
  %cpp_options -D__parallel
  ```

  in PALM‘s configuration file `.palm.config.<ci>` This is the setting in the default configuration file `.palm.config.default`.

- Additionally, the number of required cores (PEs) and the number of tasks per node (number of PEs to be used on one node) have to be provided. `-T` option is required in batch mode only.

  ```
  palmrun ... -X64 -T8 ...
  ```

- If “tasks per node“ is not an integral divisor of the total number of requested processor cores (PEs), some PEs of the „last“ node are not used (but the user probably has to pay for them).

- Using the Open-MP parallelization does not yield any advantage over using a pure domain decomposition with MPI (contrary to expectations, it mostly slows down the computational speed), but this may change on cluster systems for very large number of cores (>100000?), or in case of 1D decompositions with large number of cores.

Code parallelization

MPI communication within PALM

- MPI (message passing interface) is a portable interface for communication between PEs (FORTRAN or C library).
- All MPI calls must be within

```
CALL MPI_INIT( ierror )
...
CALL MPI_FINALIZE( ierror )
```

- MPI calls within PALM are available when using `%cpp_options -D__parallel`
- Communication is needed for
  - exchange of ghost points
  - transpositions (FFT-poisson-solver)
  - calculating statistics / global sums (e.g. for calculating horizontal averages)
  - data exchange in case of nesting
  - raytracing within the urban surface model (shading effects)
  - I/O of restart data
- Additional MPI calls are required to define the so-called virtual processor grid and to define special data types needed for more comfortable exchange of data.

Parallel I/O

- PALM writes and reads some of the input/output files in parallel, i.e. each processor writes/reads his own file. **Each file then has a different name!**
  Example: binary files for restarts are written into a subfolder of PALM‘s temporary working directory

```
$fast_io_catalog/.../BINOUT/_000000
                 .../BINOUT/_000001   etc.
```

- These files can be handled (copied) by `palmrun` using the file attribute `pe` in the configuration file `.palm.iofiles`:

```
BINOUT*  out:lnpe  restart  $fast_io_catalog/$run_identifier/RESTART _d3d
```

- In this case, filenames are interpreted as directory names. The call

```
palmrun -r example_cbl -a "... restart" ...
```

  will copy the local **directory** `BINOUT` to the user **directory**

```
.../RESTART_DATA/example_cbl_d3d
```

General comment:

- Parallel I/O on a large number of files (>1000) may cause severe file system problems (e.g. on Lustre file systems).
- **Workaround:** reduce the maximum number of parallel I/O streams (see `palmrun`-option **`-w`**), or use **`restart_data_format`** = ‘mpi‘ (PALM‘s default).

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Parallel I/O for 2D/3D data

- 2D- and 3D-data output is also written in parallel by the cores (2D: by default, 3D: generally).
- Because graphics software (`ncview`, `ncl`, `ferret`, etc.) expect the data to be in one file, these output files have to be merged to one single file after PALM has finished. This is done by the utility program `combine_plot_fields`, which is automatically executed by `palmrun` after the PALM simulation has successfully finished.
- The executable `combine_plot_fields` is created during the installation process by calling `palmbuild`.
- PALM writes 2D-data of the total domain directly into one NetCDF file (without invoking `combine_plot_fields`) if runtime parameter **`data_output_2d_on_each_pe`** `= .FALSE.` has been set.
- If you have a NetCDF4/HDF5 library which has parallel I/O support, PALM can write the 3D-data from the total domain directly into one NetCDF file without invoking `combine_plot_fields`.

  This requires setting of runtime parameter **`netcdf_data_format`** `= 5` and PALM to be compiled with cpp-options

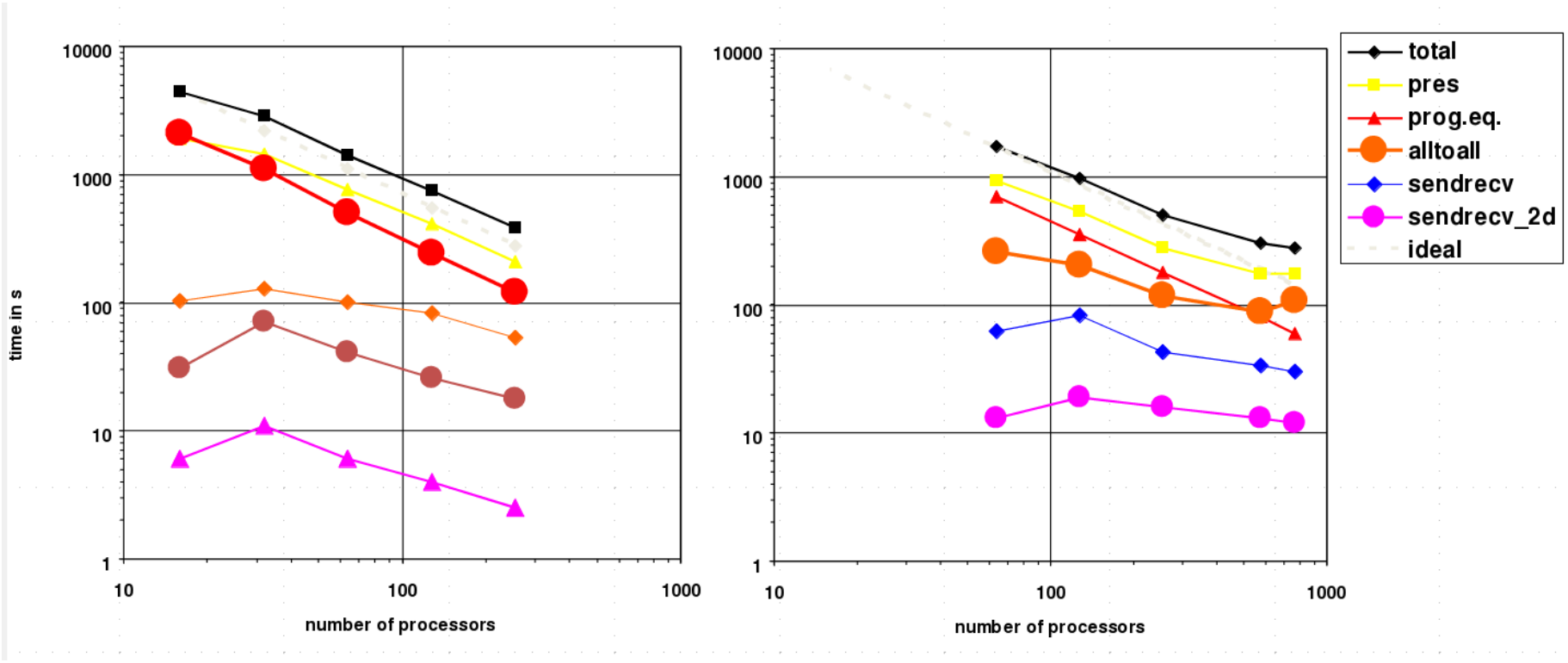
  ```
  %cpp_options ... -D__netcdf -D__netcdf4 -D__netcdf4_parallel
  ```

- **Attention:** Installation of NetCDF4/HDF5 can be very tricky.

Code parallelization

Performance examples (I)

- Simulation using 1536 * 768 * 242 grid points (~ 60 GByte)



IBM-Regatta, HLRN, Hannover (1D domain decomposition)

Sun Fire X4600, Tokyo Institute of Technology (2D domain decomposition)

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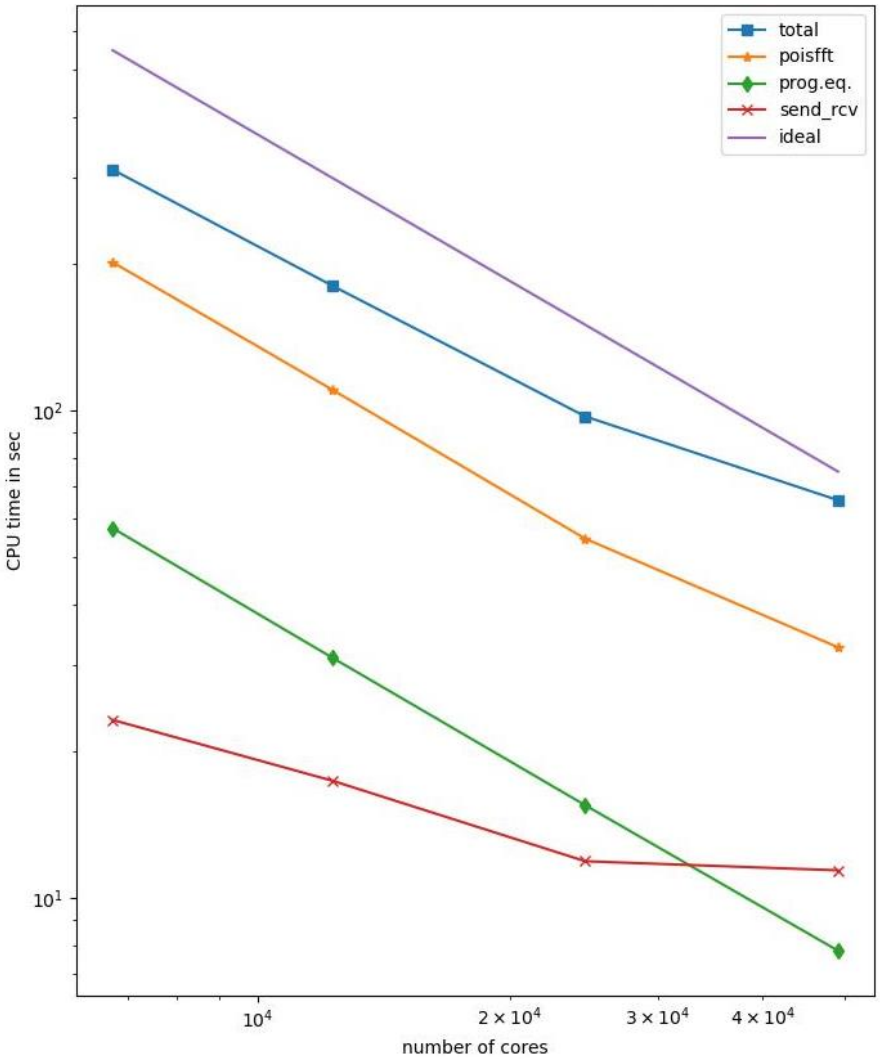
PALM - Scalability

- Simulation with 4320*4320*2160 grid points (~ 7 TByte memory)



Many parts of PALM code are already optimized for the new NEC-Aurora vector engines.

Atos/Bull Cluster, HLRN-IV, Haswell, Berlin